

● EASY

● ADVANCED MATH

● ANSWER KEY

Nonlinear functions

01

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Q1**3**

The correct answer is 3. It's given that the y-intercept of the graph shown is 0, y. The graph passes through the point 0, 3. Therefore, the value of y is 3.

Q2**B****B.**

x	1	2	3
hx	-2	1	6

Choice B is correct. It's given that $hx = x^2 - 3$. Each table gives 1, 2, and 3 as the three given values of x . Substituting 1 for x in the equation $hx = x^2 - 3$ yields $h1 = 1^2 - 3$, or $h1 = -2$. Substituting 2 for x in the equation $hx = x^2 - 3$ yields $h2 = 2^2 - 3$, or $h2 = 1$. Finally, substituting 3 for x in the equation $hx = x^2 - 3$ yields $h3 = 3^2 - 3$, or $h3 = 6$. Therefore, hx is -2 when x is 1, hx is 1 when x is 2, and hx is 6 when x is 3. Choice B is a table with these values of x and their corresponding values of hx .

Choice A is incorrect. This is a table of values for the function $hx = x + 3$, not $hx = x^2 - 3$.

Choice C is incorrect. This is a table of values for the function $hx = 2x - 3$, not $hx = x^2 - 3$.

Choice D is incorrect and may result from conceptual or calculation errors.

Q3

A

A. The object traveling at 34 meters per second has a kinetic energy of 5,202 joules.

Choice A is correct. It's given that the kinetic energy, in joules, of an object with a mass of 9 kilograms traveling at a speed of v meters per second is given by the function K , where $Kv = \frac{9}{2}v^2$. It follows that in the equation $K34 = 5,202$, 34 is the value of v , or the speed of the object, in meters per second, and 5,202 is the kinetic energy, in joules, of the object at that speed. Therefore, the best interpretation of $K34 = 5,202$ in this context is the object traveling at 34 meters per second has a kinetic energy of 5,202 joules.

Choice B is incorrect. The object traveling at 340 meters per second has a kinetic energy of 520,200 joules.

Choice C is incorrect. The object traveling at 5,202 meters per second has a kinetic energy of 121,773,618 joules.

Choice D is incorrect. The object traveling at 23,409 meters per second has a kinetic energy of 2,465,915,764.5 joules.

Q4

C

- C. 8 years after 1984, the predicted population of this country was approximately 32.91 million.
-

Choice C is correct. The function P gives the predicted population, in millions, of a certain country for the period from 1984 to 2018, where t is the number of years after 1984. Since the value of $P8$ is the value of Pt when $t = 8$, it follows that " $P8$ is approximately equal to 32.91" means that the value of Pt is approximately equal to 32.91 when $t = 8$. Therefore, the best interpretation of the statement " $P8$ is approximately equal to 32.91" is that 8 years after 1984, the predicted population of this country was approximately 32.91 million.

Choice A is incorrect. In 1984, the predicted population of this country was 24.8 million, not approximately 8 million.

Choice B is incorrect. In 1984, the predicted population of this country was 24.8 million, not approximately 32.91 million.

Choice D is incorrect. 32.91 years after 1984, the predicted population of this country was $24.81.036^{32.91}$ million, or approximately 79.42 million, not approximately 8 million.

Q5

12

The correct answer is 12. The value of $f36$ is the value of fx when $x = 36$. Substituting 36 for x in the given equation yields $f36 = 6 + \sqrt{36}$, which is equivalent to $f36 = 6 + 6$, or $f36 = 12$. Thus, the value of $f36$ is 12.

Q6**B**

B. At the end of May 2012, the estimated number of scheduled deliveries was 29.

Choice B is correct. The y-intercept of a graph in the xy -plane is the point where $x = 0$. For the given equation, the y-intercept of the graph in the xy -plane can be found by substituting 0 for x in the equation, which yields $y = -\frac{1}{4}(0)^2 + 20 + 29$, or $y = 29$. Therefore, the y-intercept of the graph is $(0, 29)$. It's given that y is the estimated number of scheduled deliveries x months after the end of May 2012. Therefore, $x = 0$ represents 0 months after the end of May 2012, or the end of May 2012. Thus, the best interpretation of the y-intercept of the graph of this equation in the xy -plane is that at the end of May 2012, the estimated number of scheduled deliveries was 29.

Choice A is incorrect and may result from conceptual errors.

Choice C is incorrect and may result from conceptual errors.

Choice D is incorrect and may result from conceptual errors.

Q7**31**

The correct answer is 31. It's given that the y-intercept of the graph of $y = x^2 + 31$ in the xy -plane is $(0, y)$. Substituting 0 for x in the given equation yields $y = 0^2 + 31$, or $y = 31$. Thus, the value of y is 31.

Q8**C****C.**

x	y
0	0
3	-12
6	0

Choice C is correct. Each point x, y on the graph represents an elapsed time x , in hours, and the corresponding ocean water level y , in feet, at a certain location based on the model. The graph shown passes through the points 0, 0, 3, -12, and 6, 0. Thus, the table in choice C gives the values of x and their corresponding values of y based on the model.

Choice A is incorrect and may result from conceptual or calculation errors.

Choice B is incorrect and may result from conceptual or calculation errors.

Choice D is incorrect and may result from conceptual or calculation errors.

Q9**C****C. 17**

Choice C is correct. It's given that $f(x) = x^3 + 9$. Substituting 2 for x in this equation yields $f(2) = 2^3 + 9$. This is equivalent to $f(2) = 8 + 9$, or $f(2) = 17$.

Choice A is incorrect. This is the value of $2 + 3 + 9$, not $2^3 + 9$.

Choice B is incorrect. This is the value of $23 + 9$, not $2^3 + 9$.

Choice D is incorrect. This is the value of $3^2 + 9$, not $2^3 + 9$.

Q10**40**

The correct answer is 40. The y-intercept of a graph in the xy -plane is the point x, y on the graph where $x = 0$. The y-intercept of the graph shown is 0, 40. Therefore, the value of y is 40.

Notes

Accept equivalent forms (e.g. $0.25 \equiv 1/4$). For grid-in items, decimal and fraction answers are both valid.

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